

Introduction to Artificial Intelligence

Artificial Intelligence (AI) refers to the simulation of human intelligence in machines that are programmed to think, reason, and learn. AI systems can perform tasks such as speech recognition, decision-making, language translation, and visual perception.

Machine Learning

Machine Learning is a subset of AI that focuses on enabling systems to learn from data. Instead of being explicitly programmed, machine learning models identify patterns and make predictions based on input data.

Natural Language Processing

Natural Language Processing (NLP) allows machines to understand and generate human language.

Applications include chatbots, virtual assistants, sentiment analysis, and document summarization.

Retrieval Augmented Generation

Retrieval Augmented Generation (RAG) combines information retrieval with text generation.

It retrieves relevant documents from a knowledge base and uses them to generate accurate answers.

Conclusion

AI is transforming industries including healthcare, finance, education, and technology.

As models become more advanced, systems like RAG improve accuracy by grounding responses in real data.